

Semantic Screening Second Read

A cold read of *Semantic Screening for Credit-Risk Feature Selection under Temporal Validation*, submitted to *Big Data Research* (elsarticle 5p, 20 pages), followed by a point-by-point check of the first referee's 34 points against the manuscript.

BASIS

LaTeX source (1,347 lines) and the rendered PDF; every figure inspected; Table 4 and the LendingClub windows re-computed.

DATE

9 September 2026

SCOPE

Part I is a new review written before consulting the first report. Part II grades each of the first referee's points. Part III orders the revision.

CONVENTION

References give section and page of the submitted PDF, e.g. §5.6 · p.10. Status colours follow the manuscript's own Okabe-Ito palette.

RECOMMENDATION

Major revision

The experimental effort is real and the reporting is candid, but the five-of-six headline claim is an out-of-time contrast between per-case maxima chosen by an unstated rule, made after the out-of-time populations had been scored in five earlier protocol versions. Two of the three datasets do not support the "temporal" framing as described. The result grid needed to check any of this is mostly withheld.

The two referee reports agree in substance. This read adds a Home Credit confound, a Stability optimism pattern, and several internal inconsistencies the first report did not raise.

26

Referee points that stand unaddressed

5

Acknowledged in the text or partly met, request unmet

3

Editorial judgement calls

0

Fully met by the submitted manuscript

34 points in the first referee's report. Two of them contain a miscount, noted in Part II; neither miscount weakens the point.

PART I Independent review

Summary of the submission

Fourteen feature-selection strategies, spanning classical filters, wrappers and embedded methods, GPT-4.1 mini semantic ranking, three LLM-classical hybrids, a CLIP-style contrastive feature representation and rank voting, are compared on three Kaggle credit datasets split into a development population and a later out-of-time (OOT) population, with penalised logistic regression at $K = 20$ and CatBoost at $K = 40$.

The headline result is that the best LLM-assisted selector attains significantly higher OOT ROC-AUC than the best classical selector in five of six dataset-model cases (+0.012 to +0.051, Holm-corrected), with Home Credit logistic

regression the exception (-0.021). Secondary results cover fold-to-fold stability, drift, cross-dataset transfer of the contrastive ranking and a rank-voting experiment on Home Credit.

What the paper does well

- The protocol is specified to a level rarely seen in this literature: seeds, package versions, both prompt templates verbatim, two explicit information contracts, budgets, the threshold rule, PSI binning.
- Paired inference on matched OOT observations with effect sizes and intervals is the right instrument at these sample sizes, and the authors explain why the five-fold rank tests were abandoned.
- The negative Home Credit result and the full protocol history are reported rather than hidden. Table 2 is the reason a referee can see what happened.
- Treating discrimination, reproducibility, drift and portability as separate outcomes is a useful framing for model governance.

Major issues ORDERED BY THREAT TO THE HEADLINE CLAIM

M1

The headline inference is not prespecified in the sense the abstract implies

ALSO RAISED BY REFEREE

§4.8 and Table 2 · pp.7-8 | §5.6 · p.10 | §6.1 · p.10

Table 2 records five protocol versions before the final one, and every row reports an out-of-time outcome: the first version's 0.772 against 0.769, the "less than 0.002" change of the contrastive correction, drift in "three of four cells", "11 of 12 canonical differences became significant", and "LLM-assisted led five of six AUC cases at the point-estimate level". After these were seen, the comparator set, the LLM's semantic inputs, the classical reference selector and the inference procedure were all changed. The "six prespecified paired comparisons" were fixed at the final version, that is, after the OOT point estimates on all three datasets were known.

Within each case the comparator is "the best classical selector of the case", but the criterion for best is never stated. If it is OOT AUC, the reported contrast is a difference between two OOT maxima over four and nine candidates, and neither the paired DeLong test nor the Holm step-down accounts for that selection. The argument in §6.1 that nine versus four makes the test conservative concerns the direction of one bias in the comparator; it does not restore the nominal level.

RESOLVE State the selection rule. If it was DEV-based, show the DEV out-of-fold AUC that produced it for every selector. If it was OOT-based, present the six contrasts as descriptive, or bootstrap the whole select-then-compare procedure over OOT resamples, or add a secondary analysis against a single classical selector fixed in advance (full-pool mRMR is the natural choice). Reword the abstract: "frozen before a single locked out-of-time evaluation" does not describe Table 2.

M2

Home Credit is not a temporal split, and the split variable lives inside the feature universe

[CONFOUND NEW IN THIS READ](#)

§5.1 · p.7 | Table 3 · p.8 | §7.4 · p.17

Time on Home Credit is the recency of the applicant's most recent previous application. DEV and OOT applications therefore come from the same calendar period and differ by a borrower characteristic, not by date. Applications without a previous application inside 600 days are dropped: 307,511 less 99,092 less 120,053 leaves 88,366 unused, which is the thin-file segment. The limitations section calls this "pseudo-temporal"; the title, abstract, §5.2 and every table call it out-of-time.

The second problem is not acknowledged. Previous-application recency and its aggregates are natural members of the 529-feature universe, and their DEV and OOT supports are disjoint by construction. A selector that picks such a feature on DEV is penalised on OOT for a reason that has nothing to do with time. The LLM prompt instructs the model to avoid "brittle operational proxies", so that penalty would fall asymmetrically on the classical selectors. The small maximum feature PSI on Home Credit suggests the winning subsets avoided these features, but the reader cannot tell which candidates were exposed.

RESOLVE List the recency-derived candidates in the universe and state whether they were removed for all selectors. Relabel the design a covariate-shift holdout throughout, discuss the dropped segment, and remove Home Credit from any sentence that claims temporal validation.

M3

The LendingClub OOT population is censored in a way that inflates its event rate

[ALSO RAISED, DATES MADE EXACT](#)

§5.1–5.2 · p.7 | Table 3 · p.8

The OOT window of –1,065 to –730 days before 2018-12-01 is loans issued January to December 2016; DEV is January 2014 to December 2015. At the 2019-05-01 snapshot no 60-month loan issued in 2016 has matured, nor has any 36-month loan issued after May 2016, nor any 60-month loan issued after mid-2014 in DEV. Keeping "resolved loans only" therefore keeps early payoffs and charge-offs for those cohorts. That is informative censoring, it acts more strongly on OOT than on DEV, and the event-rate rise from 19.54% to 23.29% is consistent with it. The sentence in §5.2 that outcomes "are resolved by the 2019-05-01 snapshot rather than censored" is incorrect for these loans. The paper's largest effect, +0.051 on LendingClub CatBoost, is measured on this population.

RESOLVE Restrict to cohorts that could mature (36-month loans issued through April 2016), or define a fixed-horizon target such as default within 24 months, and re-run. At minimum report, by term and issue month, the share of OOT loans that could not have matured.

M4

The Stability 2024 full-feature references are inherited and implausible

ALSO RAISED BY REFEREE

Table 4 footnote a · p.11 | abstract | §7.3 · p.17

A 20-feature logistic regression beating the 1,959-feature model by 0.061, and a 40-feature CatBoost beating the full-feature CatBoost by 0.068 on 1.2 million rows, is not what a depth-10, 1,500-iteration model with rsm 0.9 does unless the full universe contains strongly drifting or leaky features, or the earlier run used different data or preprocessing. Footnote a says these two models "were not refitted in the final revision", so the abstract's "exceed the full-feature reference in five of six cases" rests on two numbers from a superseded run.

RESOLVE Refit both under the final protocol. If the gap survives, show the full-feature model's DEV out-of-fold AUC and its feature PSI, because "selection as drift control" would then be a finding worth stating on its own. Otherwise remove the Stability full-feature comparison from the abstract and §7.3.

M5

Provenance of the reported precision

ALSO RAISED BY REFEREE

Table 4 and Figure 2 · p.11 | Table 2 · p.8

Adding the reported differences to the classical AUCs gives $0.802956 + 0.031444 = 0.834400$ and $0.849969 + 0.028431 = 0.878400$ for the two Stability LLM pipelines, and Figure 2's plotted coordinates use exactly those values. The caption states the differences were "computed at full precision", and Table 2 speaks of transcription and rounding errors corrected in earlier versions. Two six-decimal values ending in 00 are what a four-decimal source produces.

RESOLVE State which entries of Table 4 were computed from the final run's prediction files. Deposit the OOT score vectors (observation identifier plus score) for the twelve headline pipelines so the AUCs, differences and intervals can be recomputed by a reader.

M6

Pure LLM is not label-free on two of the three datasets

ALSO RAISED, EVIDENCE ADDED

§4.1 step 3 · pp.3-4 | §4.4 · p.4 | Table 1 · p.5 | Table 3 · p.8 | §4.7 · p.6

The LLM's candidate set on Home Credit and LendingClub is the "eligible" set of 391 and 161 features, which is what survives the information-value interval 0.01 to 0.50, a label-based screen; the prompt additionally carries training-partition moments, quantiles and missingness. §4.7 confirms it: a separate "label-free" ranking of all 529 features had to be generated for the voting experiment. Table 1's "no" in the Labels column for pure LLM is therefore wrong for two datasets, and the target-free status is only true on Stability 2024, where the 1,068-feature candidate set is a missingness screen.

RESOLVE Say plainly that the Home Credit and LendingClub candidate lists were target-screened, correct Table 1, and qualify every "label-free" statement. Also state on which partition the eligibility IV was computed; if on full DEV, every fold's out-of-fold estimate inherits a small label leak (OOT is unaffected).

M7

A single cached draw of a stochastic ranker decides four of the five wins

ALSO RAISED BY REFEREE

§4.4 · p.4 | Table 4 · p.11 | §7.4 · p.17 | §8 · p.17

Pure LLM is the winning LLM-assisted selector in Home Credit CatBoost, LendingClub LR, Stability LR and Stability CatBoost, each time from one cached response per dataset. The Stability ranking was itself the second attempt, accepted after a first response was rejected for one unknown name, so it is already the second sample of the process. The paper concedes that temperature 0 with a fixed snapshot does not guarantee identical outputs and that run-to-run variability was not measured.

RESOLVE Repeat the call, for example ten times at temperature 0 and ten at a modest temperature, and report the top-K overlap between draws and the distribution of resulting OOT AUC per case. Report the rejected first Stability response as well, after removing the unknown name.

M8

The practical recommendation and the design rationale contradict the results

ALSO RAISED, SHARPENED

Table 2 row 1 · p.8 | Table 4 · p.11 | Table 8 · p.16 | §7.1 · p.15 | §8 · p.17

Table 2 says the LLM was fixed as "a broad first-pass screener rather than the final selector" because a near-budget shortlist gave no gain. In the final results the LLM's own top-K is the winning selector in four of five wins, and on Home Credit CatBoost refining the LLM pool with mRMR lowers AUC from 0.7935 (pure LLM) to 0.7630, below full-pool mRMR (0.7668) and RFE CatBoost (0.7814). Sections 7.1 and 8 nonetheless recommend semantic screening followed by statistical refinement.

RESOLVE Report LLM then mRMR, LLM then Boruta and stable core plus LLM fill in every cell so the hybrids' full record is visible, then rewrite §7.1 and §8 to what the evidence supports: the LLM ordering itself is the effective ingredient where it wins, refinement can cost discrimination, and full-pool statistical selection wins where the universe is small enough to search.

M9

Most of the result grid is withheld

ALSO RAISED BY REFEREE

Tables 4, 5, 8 · pp.11, 13, 16

Fourteen strategies by six cases is 84 OOT AUCs; about 17 can be recovered from the tables. Random K, declared "the floor for every selector", PCA, domain rules, Boruta RF, CatBoost SHAP and LLM then Boruta never appear with a number; pure LLM's Home Credit LR value is missing; and no DEV out-of-fold AUC is given for any headline pipeline, so the reader cannot check whether the DEV-best selector was the OOT-best, which is also what M1 needs. Table 5's per-metric leaders cannot be assembled into any single model's scorecard.

RESOLVE A supplementary table with, per cell, DEV out-of-fold AUC, OOT AUC, KS, Brier, Capture@10, score PSI, realised subset size, Nogueira and Jaccard.

M10

Budget matching is asserted, not shown, for the Boruta comparators

ALSO RAISED, SETTINGS ADDED

§4.3 · p.4 | Table 4 · p.11 | §5.7 · p.10

Boruta keeps only confirmed features and the subset is "flagged as below budget" when fewer than K are confirmed. IV then Boruta is the classical comparator in three headline cases and Boruta RF is the stability leader in another. Realised subset sizes appear nowhere, and neither do Boruta's settings (number of trees, iterations, percentile, any row subsampling on 1.2 million rows) or the RFE elimination step.

RESOLVE Report the realised subset size in every cell and the wrapper settings; state whether any classical comparator ran below budget in a headline case.

M11

The voting family and the tests actually run do not match

ALSO RAISED, CAUSE IDENTIFIED

§5.6 · p.10 | §6.5 · p.15 | Table 8 · p.16 | §4.8 · p.7

The family is eight variants by three comparators, 24 Holm-corrected tests, but §6.5 says the four logistic-regression contrasts against full-pool mRMR "were not tested" against "the final full-pool reference". Read with §4.8, which records the LR mRMR reference moving from 0.7457 to 0.7699 when the RF/correlation filter was replaced by mutual-information mRMR, this indicates the voting experiment was run against the old comparator and never re-tested. The third comparator, LLM then CLIP then mRMR, has no reported AUC.

RESOLVE List the tests actually run, recompute Holm on that set, report all comparator AUCs, and say which protocol version the voting run belongs to.

M12

The contrastive branch's reproducibility claim rests on numbers outside the paper

ALSO RAISED BY REFEREE

§6.4 · pp.12–15 | Table 6 · p.13 | abstract | §7.1 · p.15

Within the paper the six CLIP cells are less reproducible than the classical leaders on the same dataset by Nogueira (0.68 to 0.75 against 0.78 to 0.79) and at best comparable by Jaccard, and they trail the headline selectors by 0.08 to 0.17 AUC. The claim that the representation "raised Jaccard by 0.32 to 0.50 while leaving discrimination unchanged" comes from an earlier Home Credit experiment whose numbers appear in one paragraph, with no table, protocol version or comparator. §7.1 also says the CLIP cells reach 0.68 to 0.75 "on the largest universe", whereas Table 6 computes them over the 60- or 100-feature pool.

RESOLVE Present the Home Credit insertion experiment properly or remove the claim from the abstract, §7.1 and §8; fix the universe statement.

M13**Every CLIP cell scores higher out of time than in development, and this is not explained****NEW IN THIS READ**

Table 7 · p.14 | §6.4 · p.12 | Table 3 · p.8

OOT AUC exceeds pooled DEV out-of-fold AUC by +0.016 to +0.045 in all six cells. The paper attributes this to composition. The Stability OOT window, 26 February to 5 October 2020, has a lower event rate (2.74% against 3.24%) and an unspecified target horizon, so outcome censoring near the end of the data window and pandemic-period interventions are both live explanations. Because DEV out-of-fold AUC is not reported for the headline pipelines, the reader cannot tell whether the same optimism sits inside the +0.031 and +0.028 Stability effects.

RESOLVE Report DEV out-of-fold and OOT AUC side by side for every headline cell and discuss the direction of the gap; say what the provider's target horizon is and whether late-2020 decisions could have been observed for it.

Minor issues

-
- m·1** The CatBoost configuration (depth 10, learning rate 0.01, L2 95, 290 observations per leaf, subsample 0.55, random strength 0.125) is evidently tuned. State on which data and features, and whether one configuration served all three datasets. **ALSO RAISED**
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- m·2** Budgets of 20 and 40 are justified by "DEV experimentation" without a curve, and §7.3's "CatBoost exceeds logistic regression in every headline case" is confounded by the unequal budgets, which §7.2 only half-concedes. **ALSO RAISED**
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- m·3** Stability is reported for the best classical selector per case only. Report Nogueira and Jaccard for every refitted selector, in particular the hybrids that win, since governance is the stated motivation. **ALSO RAISED**
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- m·4** The eligibility rule is "more than 95% missing" in §4.1 but the Stability LLM candidate set uses "at most 90% missing on the earliest training fold" (Table 3, §5.1). Reconcile. **NEW**
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- m·5** The cost paragraph in §6.5 counts 24 provider ranking calls and 48 cache hits, while §4.4 says one cached ranking per dataset. The calls presumably belong to earlier versions; say so or drop the paragraph. **NEW**
-
- m·6** Score PSI uses DEV out-of-fold scores from five fold models as the reference against OOT scores from the full-DEV refit. That is not the same scoring function, and it may be part of why the Home Credit score PSI is 0.0009 while the event rate rises a full point. Explain the choice or use full-DEV refit scores as the reference. **PARTLY RAISED**
-
- m·7** The KS-optimal operating threshold is frozen and hash-bound, but no metric at that threshold is reported. Report precision, recall and F1 at the frozen threshold or drop the machinery. **ALSO RAISED**
-
- m·8** Say that the bootstrap intervals are not multiplicity-adjusted while the p-values are. **ALSO RAISED**
-
- m·9** Define a minimal practically relevant AUC difference before counting +0.012 with a lower bound of +0.0015 as a win. **ALSO RAISED**
-

m • 10	"Fourteen strategies on three datasets" overstates coverage: the CLIP cells are on Stability only, voting on Home Credit only, and the domain rules are written against Home Credit's bureau, instalment, credit-card and previous-application tables. Add a strategy-by-dataset matrix. ALSO RAISED
m • 11	Specify the feature engineering: 675 features from LendingClub's roughly 150 columns and 1,959 from 434 Stability predictors, with aggregation functions, depths and windows. ALSO RAISED
m • 12	The domain-rules scorer adjusts priority by "leakage status". State which Home Credit features are flagged as leakage and whether they were removed from the universe for every selector. NEW
m • 13	"29 of 29 leakage checks" and "287 of 287 SHA-256 checks" are audit counts. Replace them with a sentence on what was checked and a pointer to the manifest. ALSO RAISED
m • 14	For this journal, scalability needs measurement: wall-clock and peak memory per selector and dataset under identical conditions, and complexity in the number of features and rows. The present paragraph declares itself non-comparable. ALSO RAISED
m • 15	The Stability prompt carried 1,068 feature definitions in one request and was accepted on the second attempt. Report token counts and whether the retry prompt was identical to the first. NEW
m • 16	Archive a versioned release of the repository with a DOI, including the cached rankings, the freeze manifest and the OOT score vectors of M5. NEW

Presentation

p • 1	"Appendix Appendix A" on page 6: elsarticle already prefixes appendix labels, so the word before the cross-reference must go.
p • 2	Figure 1 (p.5) justifies text inside narrow boxes, producing stretched lines and broken words ("fea- ture", "fit- ting"). Use flush-centred alignment or wider boxes.
p • 3	Figure 3 (p.13): the labels ST→ST, HC→ST and LC→ST overlap one another and the two "IV/WOE, ST" labels touch. Use leader lines or an inset for the CLIP cluster.
p • 4	Figure 4b (p.14): the diagonal is coloured at 0.5 on a scale whose maximum is 0.5 but labelled $\rho = 1$. Colour it at 1 or mask it. NEW
p • 5	The abstract's Nogueira range 0.68 to 0.75 refers to the CLIP cells without saying so; the classical selectors reach 0.96.
p • 6	Section 3 re-derives material from §2 and from §4–5 (the IV and mRMR definitions, the information-contract statement, the stability definitions); about a page can go.
p • 7	Reference 14 reads "Transactions on Machine Learning ResearchArXiv:2407.02694"; arXiv entries carry a DOI in some cases and not in others.
p • 8	The "legacy mRMR" naming history in §4.3 can shrink to the one clause Table 2 needs.

Questions to the authors

1. Was the per-case comparator chosen on DEV out-of-fold AUC or on OOT AUC?
2. On which partition was the eligibility information value computed, and were the Home Credit and LendingClub LLM prompts built from the screened set?
3. Which recency-derived features sit in the Home Credit universe, and were they eligible for every selector?
4. What were the realised subset sizes of IV then Boruta in the three headline cases where it is the comparator?
5. Which entries of Table 4 were recomputed from the final run's prediction files, and which were carried over?
6. What is the provider's target horizon on Stability 2024, and could decisions dated September and October 2020 have been observed for it?

PART II Audit of the first referee's report

Each of the 34 points, checked against the manuscript. "Stands" means the criticism is accurate and the submitted text does not address it. "Acknowledged only" means the text concedes the limitation but does not do what the referee asks. "Partly met" means part of the request is already satisfied. "Editorial" marks a judgement call where this read agrees only in part.

STANDS **ACKNOWLEDGED ONLY · PARTLY MET** **EDITORIAL** **MET** No point is fully met.

Major points

STANDS R1 · M1	Selection rule for "best of the case" never stated; both arms are OOT maxima; Holm and DeLong do not account for the selection; "9 versus 4" is not a correction. EVIDENCE §5.6 (p.10) defines the comparators only as "the best classical selector of the case" with no DEV or OOT criterion; Table 4 gives OOT AUC alone; §6.1 (p.10) offers the nine-versus-four argument; Table 2 row 5 introduces the per-case comparator at a version already reporting OOT outcomes. NEEDS State the rule; report as descriptive or use a selection-aware bootstrap; show DEV out-of-fold AUC per selector.
STANDS R2 · M1	Protocol changed after OOT results were seen; the abstract's single-lock claim is not supportable. EVIDENCE Table 2 (p.8) rows 1–5 each quote OOT outcomes from an earlier version; §4.8 (p.7) calls the revised semantic inputs "the main reason" the LLM scores changed and says the temporal design "was fixed from the outset" although Stability 2024 was added at version 5; the abstract claims one locked evaluation. NEEDS Drop the single-lock wording; describe the final run as a re-evaluation after protocol changes; show any DEV evidence behind the revision.
STANDS R3 · M9 · count refined	Only 12 of 84 selector-by-case AUCs are reported; the full grid with DEV out-of-fold AUC is needed. EVIDENCE Table 4 gives 12; Table 8 (p.16) adds Home Credit LLM then mRMR for both backbones and full-pool mRMR for CatBoost; Table 7 (p.14) adds the two native CLIP cells. About 17 of 84 are recoverable, so "roughly 70 withheld" is right. No DEV out-of-fold AUC exists for any headline pipeline. NEEDS Full grid with DEV out-of-fold AUC per cell.
STANDS R4 · M4	Stability full-feature references are implausibly weak and were not refitted. EVIDENCE Table 4 footnote a (p.11): 0.7734 and 0.8105 come from "the earlier complete run" and "were not refitted in the final revision"; the abstract and §7.3 (p.17) build the five-of-six full-feature claim on them. NEEDS Refit under the final protocol or retract the Stability full-feature claims.
STANDS R5 · M5 · confirmed	The Table 4 differences imply Stability LLM AUCs of exactly 0.834400 and 0.878400. EVIDENCE $0.802956 + 0.031444 = 0.834400$ and $0.849969 + 0.028431 = 0.878400$; Figure 2's source coordinates plot exactly these values; the caption says "computed at full precision". NEEDS State provenance; deposit score vectors.

ACKNOWLEDGED ONLY

R6 · M2

Home Credit time is prior-application recency; about 88,000 applications are dropped; label it a proxy holdout and discuss the selection effect.

EVIDENCE §5.1 (p.7) describes the DAYS_DECISION index; Table 3 leaves 88,366 of 307,511 unused; §7.4 (p.17) says "pseudo-temporal holdout", but the title, abstract, §5.2 and all tables say out-of-time, and the dropped segment is never discussed.

NEEDS Relabel throughout; discuss the selection; also address the recency-feature confound of M2, which the referee did not raise.

STANDS

R7 · M3

LendingClub's resolved-only OOT window contains 60-month 2016 loans that can only be resolved by early charge-off or prepayment; informative censoring acts differently on DEV and OOT.

EVIDENCE §5.2 (p.7) claims outcomes are "resolved ... rather than censored". The windows are January–December 2016 (OOT) and January 2014–December 2015 (DEV) against a 2019-05-01 snapshot, so no 60-month 2016 loan and no 36-month loan issued after May 2016 could have matured; the event rate rises from 19.54% to 23.29%.

NEEDS Restrict to maturable cohorts or use a fixed-horizon target; quantify the bias.

ACKNOWLEDGED ONLYR8 · M7 · count
corrected

Pure LLM wins on one cached API call per dataset; run-to-run variability was never measured.

EVIDENCE §4.4 (p.4): one cached ranking per dataset and "zero temperature do not guarantee identical outputs"; §7.4: "run-to-run variability was not measured"; §8 lists repeated runs as future work. Table 4 shows pure LLM as the winner in four of the five LLM wins, not three as the referee writes.

NEEDS Repeated calls; distribution of top-K overlap and OOT AUC.

ACKNOWLEDGED ONLY

R9

Contamination is dismissed for Stability 2024; add a neutral-identifier control.

EVIDENCE §7.4 says the definition-only contract "reduces but does not remove that exposure", so the risk is conceded rather than dismissed; the competition closed in May 2024 and the model snapshot is April 2025; no control experiment exists.

NEEDS A neutral-identifier or shuffled-description control on at least one dataset.

STANDS

R10 · M6

Were the LLM's candidate lists target-screened? If so, pure LLM is not target-free.

EVIDENCE §4.1 step 3 applies the IV interval before the hybrids; §4.4 says the LLM receives "the eligible feature names"; Table 3 (p.8) labels 391 and 161 the "LLM candidate set"; §5.1 says those counts "pass eligibility screening"; §4.7 (p.6) needed a separate "label-free" ranking of all 529 features for voting. The lists were screened.

NEEDS State it; correct Table 1; qualify every "label-free" claim.

STANDS

R11 · M10

Boruta may return fewer than K; realised subset sizes are needed to verify budget matching.

EVIDENCE §4.3 (p.4) states the below-budget rule; IV then Boruta is the comparator in LendingClub LR, LendingClub CatBoost and Stability LR (Table 4); no realised size is reported anywhere.

NEEDS Realised size per cell plus wrapper settings.

STANDSR12 · M8 · count
corrected**The recommended practice (LLM screening then statistical refinement) is not what the results support.**

EVIDENCE §7.1 (p.15) and §8 (p.17) make the recommendation; Table 4 shows pure LLM winning four cases and LLM then mRMR one (the referee writes three and one); Table 8 (p.16) shows LLM then mRMR at 0.7630 below full-pool mRMR at 0.7668 and pure LLM at 0.7935 on Home Credit CatBoost; Table 2 row 1 says the LLM was fixed as a screener "rather than the final selector".

NEEDS Rewrite §7.1 and §8; show every hybrid in every cell.

STANDS

R13 · M11

The voting family is defined as 24 tests but the LR deficits against full-pool mRMR "were not tested".

EVIDENCE §5.6 (p.10) defines $8 \times 3 = 24$; §6.5 (p.15) says the LR contrasts against "the final full-pool reference" were not tested; Table 2 records that LR reference moving from 0.7457 to 0.7699; the third comparator's AUC is never given.

NEEDS List the tests actually run; recompute Holm; report all comparator AUCs.

STANDS

R14

There is no feature-level analysis.

EVIDENCE No table of selected features, overlaps or disagreements exists; the only feature-level statement is the zero median feature PSI of the pure LLM subsets on Home Credit (§6.2, p.12).

NEEDS Selected-feature tables per cell, overlap between LLM and classical subsets, and examples of disagreement with the semantic reason the LLM gave.

Minor points

PARTLY MET

m1

CatBoost hyperparameters are clearly tuned; state on which data and features, and whether one configuration served all datasets.

EVIDENCE §5.3 (pp.7–9) lists the values; §7.4 says "one fixed configuration each", which implies a single configuration across datasets; the tuning data and feature set are not stated.

NEEDS One sentence on provenance.

STANDS

m2

No budget curve; unequal K confounds the LR versus CatBoost comparison in §7.3.

EVIDENCE §5.4 (p.9) asserts DEV experimentation with no curve; §7.3 (p.17) says CatBoost "exceeds logistic regression in every headline case"; §7.2 notes the larger budget "benefits every family" without treating it as a confound.

NEEDS A budget curve on DEV; a caveat in §7.3.

STANDS

m3

Table 6 reports stability only for the most reproducible classical selector per case.

EVIDENCE Table 6 caption (p.13); §6.3 asserts that hybrids "inherit" the reproducibility of their statistical stage without a number.

NEEDS Nogueira and Jaccard for every refitted selector.

STANDS

m4 · M12

The CLIP branch trails by 0.08 to 0.17 AUC and is less reproducible than the classical leaders; justify it or shorten §4.6 and §6.4.

EVIDENCE §6.4 (p.12) concedes the 0.08 to 0.17 gap; Table 6 gives Nogueira 0.68 to 0.75 against 0.78 to 0.96; the reproducibility benefit is cited from an earlier Home Credit experiment that is not tabulated.

NEEDS In-paper evidence for the benefit or a shorter branch.

STANDS	\$6.4 quotes results from "earlier experiments" that appear nowhere else.
m5	EVIDENCE The last paragraph of §6.4 (pp.14–15) quotes 0.7627, 0.6766, 0.5732, −0.001102, +0.000866, +0.324841, +0.498638, 0.008286 and 0.004094; none appears in any table. NEEDS A table with protocol version and comparator, or removal.
STANDS	The operating threshold is described but no threshold-dependent metric is reported.
m6	EVIDENCE §4.1 step 5 and §5.5 (p.9), which says KS is reported "as the maximum OOT separation rather than the separation at the frozen threshold"; the threshold is in the freeze list of §5.2. NEEDS Precision, recall and F1 at the frozen threshold, or drop the machinery.
STANDS	Score PSI of 0.0009 while the Home Credit event rate moves from 7.9% to 8.9% is surprising.
m7	EVIDENCE §6.2 (p.12) reports 0.000873; Table 3 gives 7.93% and 8.90%; §5.5 makes the reference bins from DEV out-of-fold scores while OOT scores come from the full-DEV refit. No explanation is offered. NEEDS Explain, check the binning and reconsider the reference (see m-6 in Part I).
PARTLY MET	Confidence intervals are bootstrap and p-values DeLong, and the intervals are not multiplicity-adjusted; say so.
m8	EVIDENCE §5.6 (p.10) states both procedures; the Table 4 caption pairs a "paired 95% confidence interval" with a "Holm-adjusted p-value" without saying the interval is unadjusted. NEEDS One sentence.
STANDS	+0.012 with a lower bound of +0.0015 is counted as a win; define a minimal practically relevant difference.
m9	EVIDENCE Table 4 Home Credit CatBoost; §7.4 says effect sizes matter more than p-values but sets no threshold. NEEDS Define one and apply it to all six cases.
STANDS	"Fourteen strategies on three datasets" overstates coverage; add a coverage matrix.
m10	EVIDENCE Table 7 (p.14) scores CLIP on Stability only (Home Credit CLIP values exist only as "earlier experiments"); §4.7 runs voting on Home Credit only; §4.3 describes the domain rules against Home Credit's auxiliary tables. NEEDS A strategy-by-dataset matrix; softer wording in the abstract.
STANDS	Feature engineering is undocumented.
m11	EVIDENCE §4.1 and §5.1 give only counts (529, 675, 1,959 from 434); §4.3 defers "remaining implementation details" to the repository. NEEDS A specification, at least in supplementary material.
EDITORIAL	Move Table 2 and §4.8 to supplementary material.
m12	EVIDENCE Both are in the main text (pp.7–8) and the Introduction lists the protocol history as contribution (iv). That history is what makes R2 visible to a reader. VIEW Agree in part. Keep a condensed history in the main text; move row-level detail and the wording about corrected transcription errors to a supplement and the response letter.

STANDS	"29 of 29" and "287 of 287" are audit counts, not evidence.
m13	EVIDENCE Last sentence of §4.6 (p.6).
	NEEDS Replace with what was checked and a manifest pointer.

STANDS	Scalability evidence is one paragraph of admittedly non-comparable timings.
m14	EVIDENCE §6.5 (p.15): "descriptive only"; "the final three-dataset run was not separately metered".
	NEEDS Controlled timings and memory per selector and dataset; complexity in features and rows.

Presentation

STANDS	"Appendix Appendix A" on page 6.
p1 · confirmed	EVIDENCE Present in the rendered PDF at §4.4, page 6.
	NEEDS Remove the literal word before the cross-reference.

STANDS	Figure 1 has stretched, hyphenated text inside boxes; Figure 3 labels collide near the CLIP points.
p2 · confirmed	EVIDENCE Both visible in the render: Figure 1 (p.5) shows "fea- ture" and "fit- ting" with stretched spacing; Figure 3 (p.13) shows the ST→ST, HC→ST and LC→ST labels overlapping and the two "IV/WOE, ST" labels touching.
	NEEDS Flush-centred box text; leader lines or an inset in Figure 3.

EDITORIAL	Section 3 largely repeats §2 and the Methods; the paper could lose two pages.
p3	EVIDENCE §3.2 re-describes IV, mRMR, RFE, Boruta and SHAP, which §4.3 defines again; §3.3 repeats the information-contract statement of §4.1; §3.4 repeats §5.5's stability definitions. Section 3 occupies about 1.3 pages.
	VIEW Agree in direction; about one page can go, not two.

STANDS	The abstract's Nogueira range refers to the CLIP cells without saying so, while classical selectors reach 0.96.
p4	EVIDENCE Abstract: "0.68 to 0.75"; Table 6: CLIP cells 0.6833 to 0.7525, classical leaders 0.7831 to 0.9588.
	NEEDS Name the cells in the abstract.

STANDS	Reference 14 reads "ResearchArXiv"; arXiv entries are inconsistently formatted.
p5	EVIDENCE The Jeong et al. entry reads "Transactions on Machine Learning ResearchArXiv:2407.02694"; the Choi, van den Oord and Zhang entries carry DOIs, the Jeong entry does not.
	NEEDS Normalise the arXiv entries.

EDITORIAL	The "legacy mRMR" naming history in §4.3 is irrelevant to readers.
p6	EVIDENCE §4.3 (p.4) carries the history; Table 2 depends on the term "RF/correlation filter".
	VIEW Agree; keep the single clause that Table 2 needs.

Where the first referee slipped. Two counts are off. Pure LLM is the winning LLM-assisted selector in four of the five wins (Home Credit CatBoost, LendingClub LR, Stability LR, Stability CatBoost), not three, in both R8 and R12. And about 17 selector-by-case OOT AUCs are recoverable from Tables 4, 7 and 8, not 12, in R3. Neither correction weakens the point; the first makes R12 stronger. Every other factual statement in the report checks out against the manuscript, including the 88,000 dropped applications (88,366), the two six-decimal values, the 24-versus-unttested voting family, and the two figure defects.

PART III Revision priorities

Ordered so that each step unblocks the ones after it. The first three decide whether the headline claim survives; the rest decide whether the paper is clean.

- 1 Release the full 84-cell grid** with DEV out-of-fold AUC, OOT AUC, secondary metrics, realised subset size, Nogueira and Jaccard, plus the OOT score vectors for the headline pipelines.
Unblocks M1, M5, M8, M9, M10, M13 and referee points R3, R5, R11, m3. Everything else is argued from this table.
- 2 State the comparator rule and re-frame the six contrasts** as descriptive, selection-aware, or against a comparator fixed in advance; rewrite the abstract's freeze language to match Table 2.
M1 and R1–R2. This is the core of the "major revision" verdict.
- 3 Refit the two Stability full-feature models** and recompute every Table 4 entry from final prediction files.
M4, M5, R4, R5. Without this the five-of-six full-feature claim cannot be kept.
- 4 Fix the LendingClub censoring** by restricting to maturable cohorts or using a fixed-horizon target, and re-run both backbones.
M3, R7. The largest reported effect depends on it.
- 5 Repeat the LLM ranking** and run a neutral-identifier control on at least one dataset; report the distribution of OOT AUC across draws.
M7, R8, R9. Four of five wins currently rest on one sample each.
- 6 Relabel Home Credit as a covariate-shift holdout**, audit recency-derived features in its universe, and correct Table 1 for pure LLM on Home Credit and LendingClub.
M2, M6, R6, R10.
- 7 Add the feature-level analysis** and report stability for every refitted selector.
R14, m3. This is what a paper about semantic screening owes its reader.
- 8 Align the narrative with the evidence:** rewrite §7.1 and §8, resolve the voting-family mismatch, either substantiate or drop the CLIP reproducibility claim, add the coverage matrix, and move the changelog detail to a supplement.
M8, M11, M12, R12, R13, m4, m5, m10, m12.
- 9 Presentation pass:** appendix reference, Figures 1, 3 and 4b, Section 3 length, reference formatting, minor inconsistencies (95% versus 90% missingness rule, call counts, threshold machinery).
p1–p9 and m4–m7 of Part I.

Status colours (vermilion, orange, grey, bluish green) are the manuscript's own Okabe-Ito palette, so the report reads in the same visual language as the figures it reviews. Page numbers refer to the submitted 20-page PDF; section numbers refer to the LaTeX source. Arithmetic checks: Table 4 differences re-added at six decimals; LendingClub windows converted from day offsets to calendar months against 2018-12-01; Home Credit counts from Table 3.